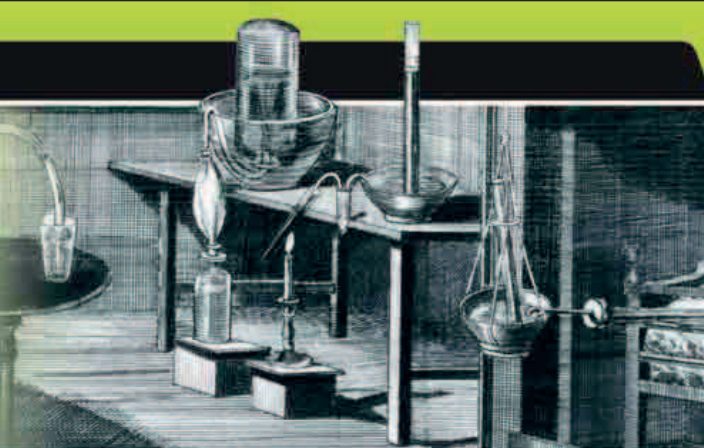


“ I HAVE... DISCOVERED AN AIR FIVE OR SIX TIMES AS GOOD AS COMMON AIR. ”

Joseph Priestley, English chemist, on his discovery of nitrous oxide, in *Experiments and Observations on Different Kinds of Air*, 1775



Joseph Priestley's laboratory housed the experimental apparatus he used for the investigation of gases.

IN 1771, GERMAN NATURALIST PETER SIMON PALLAS (1741–1811) started sending reports back from his six-year expedition to eastern Russia and Siberia, then almost as unknown to Europeans as South America. Pallas identified many **new species of plants and animals**, including what is now known as Pallas's cat.

In 1772, Swiss physicist and mathematician **Leonhard Euler** showed that the number 2,147,483,647 is a **Mersenne prime number**—a number that is one less than twice any prime number. It is named after Marin Mersenne, a French monk who had studied prime numbers in the 17th century. It was the **largest prime number** that

had been discovered for over a century. The proof involved 372 hand-calculated divisions.

Euler's fame across Europe was confirmed with the 1772 publication of his *Letters to a German Princess*. Apparently written at a rate of two per week between April 1760 and May 1763, the letters were **lessons in elementary science** for the young Prussian Princess Louise Henriette Wilhelmine of Anhalt Dessau. This 800-page work covered a wide range of scientific topics including light, color, gravity, astronomy, magnetism, optics, and more.

Also this year, Leonhard Euler's successor at the Berlin Academy, Italian-born French mathematician **Joseph Louis**

Lagrange, began an important work on mechanics, entitled *Mécanique Analytique* (*Analytical Mechanics*). From 1766 onward, Lagrange also produced a series of works on how astronomical movements could be calculated using mathematics. In particular, he looked at the **three-body problem**—how three bodies moving in space, such as the Sun, Earth, and Moon, affect each other gravitationally. This was a problem that had fascinated mathematicians for 90 years. Lagrange discovered that there are five places where a small body (such as the Moon) can remain in equilibrium with respect to two larger bodies (such as the Sun and Earth). These are now known as Lagrangian points.

In England this year, chemist **Joseph Priestley** discovered **nitrous oxide**, which he called “phlogisticated nitrous air.”



Wild cat

Pallas's cat is a small, now-endangered wild cat that lives in the mountains of central Asia. It was one of the species first identified by German naturalist Peter Pallas.

It was later used as a sedative, and became known as laughing gas because of the euphoric effects of inhaling it. The same year, Scottish physician **Daniel Rutherford** (1749–1819) **isolated nitrogen gas** in air for the first time. He called it “noxious air” because he found that mice could not survive when confined to a space filled with the gas. He also discovered that the gas did not support combustion.

“ ...NOTHING AT ALL TAKES PLACE IN THE **UNIVERSE** IN WHICH SOME RULE OF **MAXIMUM** OR **MINIMUM** DOES NOT APPEAR. ”

Leonhard Euler, Swiss mathematician, from *Methodus Inveniendi Lineas Curvas* (*Method for Finding Curved Lines*), 1744



Swiss physicist Georges-Louis LeSage using his pioneering electric telegraph.

SCIENCE HISTORY IS BESET WITH DISPUTES over who discovered things first, and one of these disputes involves the discovery of oxygen. Because of the part it played in **combustion**, and in particular the widely held phlogiston theory of combustion (see 1702–03), many chemists were investigating it in the 1770s. The first to announce its discovery was **Joseph Priestley**, who in 1775 described an experiment he conducted on August 1, 1774 in which he **created oxygen gas** (which he called “dephlogisticated air”) by focusing sunlight on mercuric oxide inside a glass tube. But in 1777, Swedish pharmacist **Carl Wilhelm Scheele** (1742–86) described how he had already produced oxygen gas by heating

DISCOVERY OF OXYGEN

Oxygen is, with nitrogen, one of two main parts of air, vital for human life and for combustion. Priestley showed that without oxygen candles will not burn and mice cannot breathe. Later, in 1777, French chemist Antoine Lavoisier went on to prove that it was oxygen involved in combustion, not phlogiston as then widely thought.

LEONHARD EULER (1707–83)

Born in Basel in Switzerland, Leonhard Euler spent most of his life in St. Petersburg and Berlin. He is credited with laying the basis of modern mathematical notation and made great advances in calculus and graph theory. His output in all areas of science was prodigious. Although in later life he was totally blind, he continued to perform calculations mentally.



1771 German naturalist **Peter Simon Pallas** reports on Siberian wildlife

1772 French mathematician **Joseph Louis Lagrange** describes the **three-body problem**

1772 Scottish physician **Daniel Rutherford** identifies **nitrogen**

1772 Swiss mathematician **Leonhard Euler** finds the largest then known **prime number**

1773 English Captain **James Cook** crosses the Antarctic Circle

1772 Swedish chemist **Carl Wilhelm Scheele** discovers **oxygen gas**

1772 English chemist **Joseph Priestley** synthesizes **nitrous oxide**

1772 **Leonhard Euler** publishes *Letters to a German Princess*

1773 French astronomer **Charles Messier** discovered a number of nebulae, drawing attention to non-stellar objects in the sky